Prof. Emiliano De Cristofaro

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Last updated: October 28, 2026

Research

- Security and Privacy Enhancing Technologies
- Trustworthy Machine Learning
- Internet Measurement, CyberSafety, Online Harms
- Google Scholar (October 2026): h-index 69; total citations 20,766

Education

- **PhD in Networked Systems**, University of California, Irvine (Advisor: Gene Tsudik)
  Dissertation: *Sharing Sensitive Information with Privacy*
  Fall 2007 – Summer 2011
- **Laurea** (5-year degree, BS + MS equivalent) in Computer Science, *summa cum laude*, top 1% in graduating class, University of Salerno, Italy
  Fall 2000 – Summer 2005

Employment

- **July 2023 – ongoing: Professor at University of California, Riverside (UCR)**

  Grants and Funding:

  - Google PSS Privacy Faculty Award, 2026 (PI, $75,000)
  - Department of Education, GAANN: Graduate Assistance in Areas of National Need (PI, $1,018,620)
  - Google Academic Research Award, On-Device Detection of Sophisticated Scams (Co-PI, $50,000)
  - Cisco Research Gift, Privacy-Friendly Collaborative Threat Mitigation (PI, $50,000)
  - DoD, AASHI: Adaptive, Autonomous and Secure Heterogeneous Integrated System of Systems (Co-I, $8,000,000)
  - UCR Committee on Research, LOGOS – A Multidisciplinary UCR initiative for modeling and improving online dialogue (PI, $3,050)

- **October 2013 – June 2023: University College London (UCL)**

  - (Full) Professor of Security and Privacy Enhancing Technologies, October 2019 – June 2023
  - Reader (~Associate Professor++), October 2017 – September 2019
  - Senior Lecturer (~Associate Professor), October 2013 – September 2017

  Roles:

  - Head of UCL's Information Security Research Group, September 2018 – August 2022
  - Director of the Academic Centre of Excellence in Cyber Security Research, January 2019 – June 2023
  - Faculty Fellow and Leader of the Privacy in Machine Learning Interest Group at the Alan Turing Institute, the UK's National Institute for Data Science and Artificial Intelligence, October 2018 – September 2021
  - Director of the MSc in Information Security, September 2015 – September 2018

  Grants and Funding:

  - Google, Privacy-Friendly Distributed Misbehavior Detection (PI, total funding $70,000)
  - Alan Turing Institute & Accenture, A Framework for Quantifying the Privacy/Utility Trade-off in Generative Model based Synthetic Data (Co-PI, total funding £150,000)
  - UKRI National Research Centre on Privacy, Harm Reduction and Adversarial Influence Online, "REPHRAIN" (Co-PI, leadership team, total funding £6,864,055; UCL funding £799,682)
  - EU H2020 MSCA-ITN, Privacy & Usability Innovative Training Network, "Privacy & Us" (PI, total funding €3,376,517; UCL funding €546,575)
  - EU H2020 MSCA-RISE, Enhancing Security and Privacy in the Social Web, "ENCASE" (PI, total funding €2,160,000; UCL funding €310,500)
  - Xerox University Affairs Committee Award and Research Councils UK IMPACT Award, "Secure Collaborative Analytics" (PI, total funding £120,000)
  - Microsoft Research & EPSRC Case Studentship, "Privacy in Distributed Learning" (PI, total funding £120,000)

- Amazon Research Awards, "Studying and Mitigating Inference Attacks on Collaborative Federated Learning" (PI, total funding $100,000)
- Alan Turing Institute & GCHQ, "Evaluating Privacy-Preserving Generative Models In The Wild" (PI, total funding £70,000)
- Google Faculty Research Award, "Enabling Progress in Genomic Research via Privacy-Preserving Data Sharing" (PI, total funding $70,000)
- Nokia Bell Labs Faculty Donation, "Privacy in Generative Models" (PI, total funding €40,000)
- Royal Society Newton Fund Grant, "Usable Cryptographic Authentication" (Co-PI, total funding £30,000)
- EPSRC iSense Exploratory Grant, "A technical framework for enabling and supporting data donors for medical research" (Co-I, total funding £120,000)
- Alan Turing Institute & GCHQ, "Privacy And Trust In The Decentralised Web" (Co-I, total funding £70,000)

- **May 2019 – May 2022: UK Information Commissioner's Office (ICO);** Technology Advisor
  - Member of the ICO's Technology Advisory Panel

- **September 2011 – September 2013: PARC (a Xerox company);** Research Scientist
  - Internal and commercial projects: secure and privacy-preserving analytics; security and privacy in healthcare and genomics; security and privacy in future Internet architectures; usable enterprise security
  - US Government projects: PI on DoE-funded National Electric Sector Cybersecurity Organization Resource program (2011–2013); co-I on several NSF, DARPA, and IARPA funded projects

- **June – September 2010: Nokia Research Center, Lausanne;** PhD Intern
  - Worked with Dr. Imad Aad and Dr. Valtteri Niemi; designed, implemented, and evaluated several protocols for security and privacy protection in smartphone applications.
  - Co-invented one US patent and co-authored one paper at IEEE PerCom

- **September – December 2009: INRIA Rhône-Alpes, Grenoble, France;** PhD Intern
  - Worked with Dr. Claude Castelluccia, discovered and demonstrated an inference attack to reconstruct search history of Google users (published at PETS)
  - Designed and implemented a system for protecting long-term privacy of published data (published at ICNP)

- **June – September 2008: NEC Europe Research Lab, Heidelberg, Germany;** PhD Intern
  - Worked with Dr. Dirk Westhoff, designed and implemented a framework for resilient data aggregation in sensor networks
  - Co-authored one paper at ACM WiSec

- **January 2006 – August 2007: University of Salerno, Italy;** Research Assistant
  - Worked with Prof. C. Blundo, Prof. G. Persiano, and Prof. V. Auletta, co-authored several academic papers
  - Research assistant on EU-funded ECRYPT and AEOLUS projects

Awards

- *IEEE S&P 2025, Distinguished Paper Award.* For the paper "The Inadequacy of Similarity-based Privacy Metrics: Privacy Attacks against 'Truly Anonymous' Synthetic Datasets," with Georgi Ganev, May 2025
- *ACM Distinguished Member (Class of 2024).* Recognized for "contributions to privacy-enhancing technologies and internet measurement," February 2025
- *ACM WebSci 2023, Best Paper Award.* For the paper "Understanding the Use of e-Prints on Reddit and 4chan's Politically Incorrect Board," with Satrio Baskoro Yudhoatmojo and Jeremy Blackburn, April 2023
- *ACM CCS 2022, Honorable Mention.* For the paper "Why So Toxic? Measuring and Triggering Toxic Behavior in Open-Domain Chatbots," with Wai Man Si, Michael Backes, Jeremy Blackburn, Gianluca Stringhini, Savvas Zannettou, and Yang Zhang, November 2022
- *ACM CSCW 2021, Honorable Mention.* For the paper "Do Platform Migrations Compromise Content Moderation? Evidence from r/The_Donald and r/Incels," with Manoel Horta Ribeiro, Shagun Jhaver, Savvas Zannettou, Jeremy Blackburn, Gianluca Stringhini, and Robert West, October 2021
- *ACM CSCW 2021, Impact Recognition Award.* For the paper "I'm a Professor, which isn't usually a dangerous job: Internet-Facilitated Harassment and its Impact on Researchers," with Periwinkle Doerfler, Andrea Forte, Gianluca Stringhini, Jeremy Blackburn, and Damon McCoy, October 2021
- *INRIA-CNIL Privacy Protection Award 2019, Runner-Up.* For the paper "Knock Knock, Who's There? Membership Inference on Aggregate Location Data," with Apostolos Pyrgelis and Carmela Troncoso, 2019
- *ACM CyberSafety 2019, Best Paper Award.* For the paper "Disinformation Warfare: Understanding State-Sponsored Trolls on Twitter and Their Influence on the Web," with Savvas Zannettou, Tristan Caulfield, Michael Sirivianos, Gianluca Stringhini, and Jeremy Blackburn, May 2019
- *ACM IMC 2018, Distinguished Paper Award.* For the paper "On the Origins of Memes by Means of Fringe Web Communities," with Savvas Zannettou, Tristan Caulfield, Jeremy Blackburn, Michael Sirivianos, Gianluca Stringhini, and Guillermo Suarez-Tangil, October 2018

- *NDSS 2018, Distinguished Paper Award.* For the paper "Knock Knock, Who's There? Membership Inference on Aggregate Location Data," with Apostolos Pyrgelis and Carmela Troncoso, February 2018
- *CSAW Europe 2017 Applied Research Award, 2nd Place (out of 128 papers).* For the paper "MaMaDroid: Detecting Android Malware by Building Markov Chains of Behavioral Models," with Enrico Mariconti, Lucky Onwuzurike, Panagiotis Andriotis, Gordon Ross, and Gianluca Stringhini, November 2017
- *5th Data Protection by Design Award.* Award issued by the Catalan Data Protection authority and the Parliament of Catalonia, for work with Luca Melis and George Danezis, June 2017
- *AAAI ICWSM 2017, Best Paper Runner-Up.* For the paper "Kek, Cucks, and God Emperor Trump: A Measurement Study of 4chan's Politically Incorrect Forum and Its Effects on the Web," with Gabriel Emile Hine, Jeremiah Onaolapo, Nicolas Kourtellis, Ilias Leontiadis, Riginos Samaras, Gianluca Stringhini, and Jeremy Blackburn, May 2017
- *ACM IMC 2014, Best Paper Runner-Up.* For the paper "Censorship in the Wild: Analyzing Internet Filtering in Syria," with Abdelberi Chaabane, Terence Chen, Mathieu Cunche, Arik Friedman, and Mohamed Ali Kaafar, November 2014
- *PARC Excellence Award.* Outstanding research performance in 2012, only 1 awarded per group
- *UC Irvine's Dissertation Fellowship.* Fall 2010 ($10,000), only 1 awarded per department
- *UC Irvine's Dean's Fellowship.* 4-year PhD financial support (2007–2011), only 5 awarded per department

Selected Publications*

1. M.S.M.S. Annamalai, E. De Cristofaro, P. Kairouz
   CLIOPATRA: Extracting Private Information from LLM Insights
   Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)
   Oral Presentation
2. G. Ganev, M.R. Nazari, R. Davison, A.F. Dizche, X. Wu, R. Abbey, J.G. Silva, E. De Cristofaro
   SMOTE and Mirrors: Exposing Privacy Leakage from Synthetic Minority Oversampling
   14th International Conference on Learning Representations (ICLR 2026)
3. M.S.M.S. Annamalai, B. Balle, J. Hayes, E. De Cristofaro
   To Shuffle or not to Shuffle: Auditing DP-SGD with Shuffling
   33rd Network and Distributed System Security Symposium (NDSS 2026)
4. A. Galeazzi, P. Paudel, M. Conti, E. De Cristofaro, G. Stringhini
   Revealing The Secret Power: How Algorithms Can Influence Content Visibility on Twitter/X
   33rd Network and Distributed System Security Symposium (NDSS 2026)
5. G. Ganev, M.S.M.S. Annamalai, S. Mahiou, E. De Cristofaro
   The Importance of Being Discrete: Measuring the Impact of Discretization in End-to-End Differentially Private Synthetic Data
   32nd ACM Conference on Computer and Communications Security (ACM CCS 2025)
6. G. Ganev and E. De Cristofaro
   The Inadequacy of Similarity-based Privacy Metrics: Privacy Attacks against 'Truly Anonymous' Synthetic Datasets
   46th IEEE Symposium on Security and Privacy (S&P 2025)
   Distinguished Paper Award
7. M.S.M.S. Annamalai, I. Bilogrevic, E. De Cristofaro
   Beyond the Crawl: Unmasking Browser Fingerprinting in Real User Interactions
   34th The Web Conference (WWW 2025)
8. M.S.M.S. Annamalai and E. De Cristofaro
   Nearly Tight Black-Box Auditing of Differentially Private Machine Learning
   Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024)
9. G. Ganev, K. Xu, E. De Cristofaro
   Graphical vs. Deep Generative Models: Measuring the Impact of Differentially Private Mechanisms and Budgets on Utility
   31st ACM Conference on Computer and Communications Security (CCS 2024)
10. M.S.M.S. Annamalai, G. Ganev, E. De Cristofaro
    "What do you want from theory alone?" Experimenting with Tight Auditing of Differentially Private Synthetic Data Generation
    33rd USENIX Security Symposium (USENIX Security 2024)
11. A. Efstratiou, M. Efstratiou, S. Yudhoatmojo, J. Blackburn, E. De Cristofaro
    "Here's Your Evidence": False Consensus in Public Twitter Discussions of COVID-19 Science
    27th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2024)

*Listing: 1) conferences listed as the "most selective" in Computer Science according to csrankings.org (36), 2) top specialized venues in computational social science, i.e., ICWSM and CSCW (20), and 3) the top specialized venue on privacy technologies, i.e., PETS (5).

12. M.S.M.S. Annamalai, I. Bilogrevic, E. De Cristofaro
FP-Fed: Privacy-Preserving Federated Detection of Browser Fingerprinting
31st Network and Distributed System Security Symposium (NDSS 2024)
13. M. Naseri, Y. Han, E. De Cristofaro
BadVFL: Backdoor Attacks in Vertical Federated Learning
45th IEEE Symposium on Security and Privacy (S&P 2024)
14. M.H. Saeed, K. Papadamou, J. Blackburn, E. De Cristofaro, G. Stringhini
TUBERAIDER: Attributing Coordinated Hate Attacks on YouTube Videos to their Source Communities
18th International AAAI Conference on Web and Social Media (ICWSM 2024)
15. J. Patel, P. Paudel, E. De Cristofaro, G. Stringhini, J. Blackburn
iDRAMA-Scored-2024: A Dataset of the Scored Social Media Platform from 2020 to 2023
18th International AAAI Conference on Web and Social Media, Dataset Paper (ICWSM 2024)
16. P. Paudel, J. Blackburn, E. De Cristofaro, S. Zannettou, G. Stringhini
LAMBRETTA: Learning to Rank for Twitter Soft Moderation
44th IEEE Symposium on Security & Privacy (S&P 2023)
17. A. Efstratiou, J. Blackburn, T. Caulfield, G. Stringhini, S. Zannettou, E. De Cristofaro
Non-Polar Opposites: Analyzing the Relationship Between Echo Chambers and Hostile Intergroup Interactions on Reddit
17th International AAAI Conference on Web and Social Media (ICWSM 2023)
18. M. Naseri, Yufei Han, E. Mariconti, Y. Shen, G. Stringhini, E. De Cristofaro
CERBERUS: Exploring Federated Prediction of Security Events
29th ACM Conference on Computer and Communications Security (CCS 2022)
19. W.M. Si, M. Backes, J. Blackburn, E. De Cristofaro, G. Stringhini, S. Zannettou, Yang Zhang
Why So Toxic? Measuring and Triggering Toxic Behavior in Open-Domain Chatbots
29th ACM Conference on Computer and Communications Security (CCS 2022)
Honorable Mention
20. A. Efstratiou and E. De Cristofaro
Adherence to Misinformation on Social Media Through Socio-Cognitive and Group-Based Processes
25th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2022)
21. G. Ganev, B. Oprisanu, E. De Cristofaro
Robin Hood and Matthew Effects – Differential Privacy Has Disparate Impact on Synthetic Data
39th International Conference on Machine Learning (ICML 2022)
22. Y. Liu, R. Wen, X. He, A. Salem, Z. Zhang, M. Backes, E. De Cristofaro, M. Fritz, Y. Zhang
ML-Doctor: Holistic Risk Assessment of Inference Attacks Against Machine Learning Models
31st USENIX Security Symposium (USENIX Security 2022)
23. H.B. Zia, A. Raman, I. Castro, I. Anaobi, E. De Cristofaro, N. Sastry, G. Tyson
Toxicity in the Decentralized Web and the Potential for Model Sharing
ACM International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS 2022)
24. M.H. Saeed, S. Ali, J. Blackburn, E. De Cristofaro, S. Zannettou, G. Stringhini
TROLLMAGNIFIER: Detecting State-Sponsored Troll Accounts on Reddit
43rd IEEE Symposium on Security & Privacy (S&P 2022)
25. M. Naseri, J. Hayes, E. De Cristofaro
Local and Central Differential Privacy for Robustness and Privacy in Federated Learning
29th Network and Distributed System Security Symposium (NDSS 2022)
26. B. Oprisanu, G. Ganev, E. De Cristofaro
On Utility and Privacy in Synthetic Genomic Data
29th Network and Distributed System Security Symposium (NDSS 2022)
27. A. Papasavva, M. Aliapoulios, C. Ballard, E. De Cristofaro, G. Stringhini, S. Zannettou, J. Blackburn
The Gospel According to Q: Understanding the QAnon Conspiracy from the Perspective of Canonical Information
16th International AAAI Conference on Web and Social Media (ICWSM 2022)
28. K. Papadamou, S. Zannettou, J. Blackburn, E. De Cristofaro, G. Stringhini, M. Sirivianos
"It is just a flu": Assessing the Effect of Watch History on YouTube's Pseudoscientific Video Recommendations
16th International AAAI Conference on Web and Social Media (ICWSM 2022)
29. P. Doerfler, A. Forte, E. De Cristofaro, G. Stringhini, J. Blackburn, D. McCoy
"I'm a Professor, which isn't usually a dangerous job": Internet-Facilitated Harassment and its Impact on Researchers
24th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2021)
Impact Recognition Award
30. M. Horta Ribeiro, S. Jhaver, S. Zannettou, J. Blackburn, E. De Cristofaro, G. Stringhini, R. West
Do Platform Migrations Compromise Content Moderation? Evidence from r/The_Donald and r/Incels
24th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2021)

Honorable Mention
Media coverage: WaPo, El País, The Brink, Wired

31. K. Papadamou, S. Zannettou, J. Blackburn, E. De Cristofaro, G. Stringhini, M. Sirivianos
"How over is it?" Understanding the Incel Community on YouTube
24th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2021)
32. C. Ling, I. AbuHilal, J. Blackburn, E. De Cristofaro, S. Zannettou, G. Stringhini
Dissecting the Meme Magic: Understanding Indicators of Virality in Image Memes
24th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2021)
Media coverage: Input Mag
33. A. Papasavva, J. Blackburn, G. Stringhini, S. Zannettou, E. De Cristofaro
"Is it a Qoincidence?": An Exploratory Study of QAnon on Voat
30th The Web Conference (WWW 2021)
Media coverage: WaPo, El País, The Brink, Wired
34. Max Aliapoulios, Emmi Bevensee, J. Blackburn, E. De Cristofaro, G. Stringhini, S. Zannettou
A Large Open Dataset from the Parler Social Network
15th International AAAI Conference on Web and Social Media (ICWSM 2021)
Media coverage: USA Today, The Conversation, Business Insider-1, Business Insider-2
35. Y. Wang, F. Tahmasbi, J. Blackburn, B. Bradlyn, E. De Cristofaro, D. Magerman, S. Zannettou, G. Stringhini
Understanding the Use of Fauxtography on Social Media
15th International AAAI Conference on Web and Social Media (ICWSM 2021)
36. M. Horta Ribeiro, J. Blackburn, B. Bradlyn, E. De Cristofaro, G. Stringhini, S. Long, S. Greenberg, S. Zannettou
The Evolution of the Manosphere Across the Web
15th International AAAI Conference on Web and Social Media (ICWSM 2021)
Media coverage: Newsweek, MIT Tech, Daily Mail
37. A. Pyrgelis, C. Troncoso, E. De Cristofaro
Measuring Membership Privacy on Aggregate Location Time-Series
ACM International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS 2020)
38. A. Papasavva, S. Zannettou, E. De Cristofaro, G. Stringhini, J. Blackburn
Raiders of the Lost Kek: 3.5 Years of Augmented 4chan Posts from the Politically Incorrect Board
14th International AAAI Conference on Web and Social Media (ICWSM 2020)
39. S. Zannettou, B. Bradlyn, T. Caulfield, E. De Cristofaro, G. Stringhini, J. Blackburn
Characterizing the Use of Images in State-Sponsored Information Warfare Operations by Russian Trolls on Twitter
14th International AAAI Conference on Web and Social Media (ICWSM 2020)
40. A. Mittos, S. Zannettou, J. Blackburn, E. De Cristofaro
"And We Will Fight For Our Race!" A Measurement Study of Genetic Testing Conversations on Reddit and 4chan
14th International AAAI Conference on Web and Social Media (ICWSM 2020)
41. A. Raman, S. Joglekar, E. De Cristofaro, N. Sastry, G. Tyson
Challenges in the Decentralized Web: The Mastodon Case
19th ACM Internet Measurement Conference (IMC 2019)
42. E. Mariconti, G. Suarez-Tangil, J. Blackburn, E. De Cristofaro, N. Kourtellis, I. Leontiadis, J. Luque Serrano, G. Stringhini
"You Know What to Do": Proactive Detection of YouTube Videos Targeted by Coordinated Hate Attacks
22nd ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2019)
43. L. Melis, C. Song, E. De Cristofaro, V. Shmatikov
Exploiting Unintended Feature Leakage in Collaborative Learning
40th IEEE Symposium on Security & Privacy (S&P 2019)
44. V. Primault, V. Lampos, I. Cox, E. De Cristofaro
Privacy-Preserving Crowd-Sourcing of Web Searches with Private Data Donor
28th The World Wide Web Conference (WWW 2019)
45. Jamie Hayes, L. Melis, G. Danezis, E. De Cristofaro
LOGAN: Membership Inference Attacks Against Generative Models
Proceedings on Privacy Enhancing Technologies, Vol. 2019, Issue 1 (PoPETS 2019)
46. A. Mittos, Bradley Malin, E. De Cristofaro
Systematizing Genome Privacy Research: A Privacy-Enhancing Technologies Perspective
Proceedings on Privacy Enhancing Technologies, Vol. 2019, Issue 1 (PoPETS 2019)
47. S. Zannettou, T. Caulfield, J. Blackburn, E. De Cristofaro, M. Sirivianos, G. Stringhini, G. Suarez-Tangil
On the Origins of Memes by Means of Fringe Web Communities
18th ACM Internet Measurement Conference (IMC 2018)
Distinguished Paper Award

48. A. Pyrgelis, C. Troncoso, E. De Cristofaro
Knock Knock, Who's There? Membership Inference on Aggregate Location Data
25th Network and Distributed System Security Symposium (NDSS 2018)
Distinguished Paper Award; INRIA-CNIL Privacy Protection Award Runner-Up
49. S. Zannettou, J. Blackburn, E. De Cristofaro, M. Sirivianos, G. Stringhini
Understanding Web Archiving Services and Their (Mis)Use on Social Media
12th International AAAI Conference on Web and Social Media (ICWSM 2018)
50. B. Oprisanu and E. De Cristofaro
AnoniMME: Bringing Anonymity to the Matchmaker Exchange Platform for Rare Disease Gene Discovery
26th ISCB Conference on Intelligent Systems for Molecular Biology (ISMB 2018)
51. S. Zannettou, T. Caulfield, E. De Cristofaro, N. Kourtellis, I. Leontiadis, M. Sirivianos, G. Stringhini, J. Blackburn
The Web Centipede: Understanding How Web Communities Influence Each Other Through the Lens of Mainstream and Alternative News Sources
17th ACM Internet Measurement Conference (IMC 2017)
52. Apostolos Pyrgelis, Carmela Troncoso, E. De Cristofaro
What Does The Crowd Say About You? Evaluating Aggregation-based Location Privacy
17th Privacy Enhancing Technologies Symposium (PETS 2017)
53. G. Hine, J. Onaolapo, E. De Cristofaro, N. Kourtellis, I. Leontiadis, R. Samaras, G. Stringhini, J. Blackburn
Kek, Cucks, and God Emperor Trump: A Measurement Study of 4chan's Politically Incorrect Forum and Its Effects on the Web
11th International AAAI Conference on Web and Social Media (ICWSM 2017)
Best Paper Runner-Up
54. E. Mariconti, L. Onwuzurike, P. Andriotis, E. De Cristofaro, G. Ross, G. Stringhini
MaMaDroid: Detecting Android Malware by Building Markov Chains of Behavioral Models
24th Network and Distributed System Security Symposium (NDSS 2017)
CSAW Europe Applied Research Award, 2nd Place
55. L. Melis, G. Danezis, E. De Cristofaro
Efficient Private Statistics with Succinct Sketches
23rd Network and Distributed System Security Symposium (NDSS 2016)
56. A. Chaabane, T. Chen, M. Cunche, E. De Cristofaro, A. Friedman, M.A. Kaafar
Censorship in the Wild: Analyzing Internet Filtering in Syria
14th ACM Internet Measurement Conference (IMC 2014)
Best Paper Runner-Up
57. E. De Cristofaro, A. Friedman, G. Jourjon, M.A. Kaafar, M.Z. Shafiq
Paying for Likes? Understanding Facebook Like Fraud Using Honeypots
14th ACM Internet Measurement Conference (IMC 2014)
58. E. De Cristofaro, C. Soriente, G. Tsudik, A. Williams
Hummingbird: Privacy at the time of Twitter
IEEE Symposium on Security and Privacy (S&P 2012)
59. P. Baldi, R. Baronio, E. De Cristofaro, P. Gasti, G. Tsudik
Countering GATTACA: Efficient and Secure Testing of Fully Sequenced Human Genomes
ACM Conference on Computer and Communications Security (CCS 2011)
60. C. Castelluccia, E. De Cristofaro, D. Perito
Private Information Disclosure from Web Searches
Privacy Enhancing Technologies Symposium (PETS 2010)
Also appeared as a poster at IEEE Symposium on Security and Privacy (S&P 2010)
61. E. De Cristofaro, S. Jarecki, J. Kim, G. Tsudik
Privacy-preserving Policy-based Information Transfer
Privacy Enhancing Technologies Symposium (PETS 2009)

Other Publications

1. G. Ganev, E. De Cristofaro
Rethinking Anonymity Claims in Synthetic Data Generation: A Model-Centric Privacy Attack Perspective
25th ACM CCS Workshop on Privacy in the Electronic Society (WPES 2026)
2. B. Treves, M. Treves, M. Faloutsos, J. Blackburn, E. De Cristofaro
ASTRO: Defining and Identifying Hostile Behavior on GitHub
7th International Workshop on Cyber Social Threats (CySoc 2026, co-located with ICWSM 2026)
3. P. Gaba, E. De Cristofaro
Group-Differentiated Discourse on Generative AI in High School Education: A Case Study of Reddit Communities
7th International Workshop on Cyber Social Threats (CySoc 2026, co-located with ICWSM 2026)

4. M.S.M.S. Annamalai, B. Balle, J. Hayes, G. Kaissis, E. De Cristofaro
SoK: The Hitchhiker's Guide to Efficient, End-to-End, and Tight DP Auditing
4th IEEE Secure and Trustworthy Machine Learning Conference (IEEE SaTML 2026)
5. S. Chennoufi, Y. Han, G. Blanc, E. De Cristofaro, C. Kiennert
PROTEAN: Federated Intrusion Detection in Non-IID Environments through Prototype-Based Knowledge Sharing
30th European Symposium on Research in Computer Security (ESORICS 2025)
6. B. Treves, E. De Cristofaro, M. Faloutsos, Y. Dong
VIKI: Systematic Cross-Platform Profile Inference of Tech Users
17th ACM Web Science Conference (WebSci 2025)
7. G. Ganev, M.S.M.S. Annamalai, S. Mahiou, E. De Cristofaro
Understanding the Impact of Data Domain Extraction on Synthetic Data Privacy
Will Synthetic Data Finally Solve the Data Access Problem? (ICLR 2025 Workshop)
8. G. Ganev, M.S.M.S. Annamalai, E. De Cristofaro
The Elusive Pursuit of Reproducing PATE-GAN: Benchmarking, Auditing, Debugging
Transactions on Machine Learning Research (TMLR), Feb 2025
9. E. De Cristofaro
Synthetic Data: Methods, Use Cases, and Risks
IEEE Security and Privacy Magazine – Special Issue on Synthetic Realities, 2024
10. S.B. Yudhoatmojo, E. De Cristofaro, J. Blackburn
Understanding the Use of e-Prints on Reddit and 4chan's Politically Incorrect Board
15th ACM Web Science Conference 2023 (WebSci 2023)
Best Paper Award
11. U. Balci, C. Ling, E. De Cristofaro, M. Squire, G. Stringhini, J. Blackburn
Beyond Fish and Bicycles: Exploring the Varieties of Online Women's Ideological Spaces
15th ACM Web Science Conference 2023 (WebSci 2023)
12. C. Ling, J. Blackburn, E. De Cristofaro, G. Stringhini
Slapping Cats, Bopping Heads, and Oreo Shakes: Understanding Indicators of Virality in TikTok Short Videos
14th ACM Web Science Conference (ACM WebSci 2022)
13. Y. Wang, S. Zannettou, J. Blackburn, B. Bradlyn, E. De Cristofaro, G. Stringhini
A Multi-Platform Analysis of Political News Discussion and Sharing on Web Communities
2021 IEEE International Conference on Big Data (IEEE BigData 2021)
14. A.I. Hassan, A. Raman, I. Castro, H.B. Zia, E. De Cristofaro, N. Sastry, G. Tyson
Exploring Content Moderation in the Decentralised Web: The Pleroma Case
17th International Conference on emerging Networking EXperiments and Technologies (CoNEXT 2021)
15. E. De Cristofaro
A Critical Overview of Privacy in Machine Learning
IEEE Security & Privacy Magazine, Volume 19, Issue 4, July-August 2021
16. S. Ali, M.H. Saeed, E. Aldreabi, J. Blackburn, E. De Cristofaro, S. Zannettou, G. Stringhini
Understanding the Effect of Deplatforming on Social Networks
13th ACM Web Science Conference (ACM WebSci 2021)
17. A. Mittos, S. Zannettou, J. Blackburn, E. De Cristofaro
Analyzing Genetic Testing Discourse on the Web Through the Lens of Twitter, Reddit, and 4chan
ACM Transactions on the Web (TWEB), Volume 13, Issue 4, August 2020
18. M. Wixey, S. Johnson, E. De Cristofaro
On the Feasibility of Acoustic Attacks Using Commodity Smart Devices
IEEE Workshop on the Internet of Safe Things (co-located with IEEE S&P 2020)
19. B. Oprisanu, C. Dessimoz, E. De Cristofaro
How Much Does GenoGuard Really Guard? An Empirical Analysis of Long-Term Security for Genomic Data
18th ACM CCS Workshop on Privacy in the Electronic Society (WPES 2019)
20. H.J. Asghar, E. De Cristofaro, G. Jourjon, D. Kaafar, L. Mathy, L. Melis, C. Russell, M. Yu
Fast Privacy-Preserving Network Function Outsourcing
Computer Networks, Vol. 163, 2019
21. D. Chatzakou, I. Leontiadis, J. Blackburn, E. De Cristofaro, G. Stringhini, A. Vakali, N. Kourtellis
Detecting Cyberbullying and Cyberaggression in Social Media
ACM Transactions on the Web (TWEB), Vol. 13, No. 3, 2019
22. S. Zannettou, T. Caulfield, E. De Cristofaro, M. Sirivianos, G. Stringhini, J. Blackburn
Disinformation Warfare: Understanding State-Sponsored Trolls on Twitter and Their Influence on the Web
4th Workshop on Computational Methods in CyberSafety, Online Harassment and Misinformation (ACM CyberSafety 2019)
Best Paper Award

23. L. Onwuzurike, E. Mariconti, P. Andriotis, E. De Cristofaro, G. Ross, G. Stringhini
MaMaDroid: Detecting Android Malware by Building Markov Chains of Behavioral Models (Extended Version)
ACM Transactions on Privacy and Security (ACM TOPS), Volume 22, Issue 2, April 2019
24. L. Melis, Apostolos Pyrgelis, E. De Cristofaro
On Collaborative Predictive Blacklisting
ACM SIGCOMM's Computer Communication Review (CCR), Vol. 48, No. 5, Oct. 2018
25. J. Echeverria, E. De Cristofaro, N. Kourtellis, I. Leontiadis, G. Stringhini, Shi Zhou
LOBO — Evaluation of Generalization Deficiencies in Twitter Bot Classifiers
2018 Annual Computer Security Applications Conference (ACSAC 2018)
26. G. Acs, L. Melis, C. Castelluccia, E. De Cristofaro
Differentially Private Mixture of Generative Neural Networks (Extended Version)
IEEE Transactions on Knowledge and Data Engineering (TKDE 2018)
27. N. Kotonya, P. De Cristofaro, E. De Cristofaro
Of Wines and Reviews: Measuring and Modeling the Vivino Wine Social Network
IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2018)
28. L. Onwuzurike, Mario Almeida, E. Mariconti, J. Blackburn, G. Stringhini, E. De Cristofaro
A Family of Droids – Android Malware Detection via Behavioral Modeling: Static vs Dynamic Analysis
16th Annual Conference on Privacy, Security and Trust (PST 2018)
29. A. Cerulli, E. De Cristofaro, C. Soriente
Nothing Refreshes Like a RePSI: Reactive Private Set Intersection
16th International Conference on Applied Cryptography and Network Security (ACNS 2018)
30. A. Massimo Perillo and E. De Cristofaro
PAPEETE: Private, Authorized, and Fast Personal Genomic Testing
15th International Conference on Security and Cryptography (SECRYPT 2018)
31. S. Zannettou, B. Bradlyn, E. De Cristofaro, H. Kwak, M. Sirivianos, G. Stringhini, J. Blackburn
What is Gab? A Bastion of Free Speech or an Alt-Right Echo Chamber?
3rd WWW Workshop on Computational Methods for CyberSafety (CyberSafety 2018)
32. G. Acs, L. Melis, C. Castelluccia, E. De Cristofaro
Differentially Private Mixture of Generative Neural Networks
17th IEEE International Conference on Data Mining series (ICDM 2017)
33. M. Ikram, L. Onwuzurike, S. Farooqi, E. De Cristofaro, A. Friedman, G. Jourjon, D. Kaafar, M.Z. Shafiq
Measuring, Characterizing, and Detecting Facebook Like Farms
ACM Transactions on Privacy and Security (ACM TOPS 2017)
34. D. Chatzakou, N. Kourtellis, J. Blackburn, E. De Cristofaro, G. Stringhini, A. Vakali
Hate is not binary: Studying abusive behavior of #GamerGate on Twitter
28th ACM Conference on Hypertext and Social Media (ACM HyperText 2017)
35. D. Chatzakou, N. Kourtellis, J. Blackburn, E. De Cristofaro, G. Stringhini, A. Vakali
Mean Birds: Detecting Aggression and Bullying on Twitter
9th International ACM Web Science Conference (ACM WebSci 2017)
36. S. Farooqi, M. Ikram, E. De Cristofaro, A. Friedman, G. Jourjon, M.A. Kaafar, M.Z. Shafiq, F. Zaffar
Characterizing Key Stakeholders in an Online Black-Hat Marketplace
12th IEEE/APWG Symposium on Electronic Crime Research (eCrime 2017)
37. D. Chatzakou, N. Kourtellis, J. Blackburn, E. De Cristofaro, G. Stringhini, A. Vakali
Measuring #GamerGate: A Tale of Hate, Sexism, and Bullying
2nd WWW Workshop on Computational Methods for CyberSafety (CyberSafety 2017)
38. S. Sajadmanesh, S. Jafarzadeh, S.A. Ossia, H.R. Rabiee, H. Haddadi, Y. Mejova, M. Musolesi, E. De Cristofaro, G. Stringhini
Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web
26th International World Wide Web Conference Web Science Track (WWW 2017)
39. H. Haddadi, R. Nithyanand, S. Khattak, M. Javed, N. Vallina-Rodriguez, M. Falahrastegar, J.E. Powles, E. De Cristofaro, S.J. Murdoch
The Adblocking Tug-of-War
USENIX ;login: Magazine, Winter 2016, Vol. 41, No. 4 (;login: 2016)
40. Apostolos Pyrgelis, E. De Cristofaro, G. Ross
Privacy-Friendly Mobility Analytics using Aggregate Location Data
24th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL 2016)
41. E. De Cristofaro, K. Liang, Y. Zhang
Privacy-Preserving Genetic Relatedness Test
3rd International Workshop on Genome Privacy and Security (GenoPri 2016)

42. R. Nithyanand, S. Khattak, M. Javed, N. Vallina-Rodriguez, M. Falahrastegar, J. E. Powles, E. De Cristofaro, H. Haddadi, S.J. Murdoch
Ad-Blocking and Counter Blocking: A Slice of the Arms Races
6th USENIX Workshop on Free and Open Communications on the Internet (FOCI 2016)
43. H. Asghar, L. Melis, C. Soldani, E. De Cristofaro, M.A. Kaafar, L. Mathy
SplitBox: Toward Efficient Private Network Function Virtualization
3rd ACM SIGCOMM Workshop on Hot Topics in Middleboxes and Network Function Virtualization (HotMiddleBox 2016)
44. L. Melis, H. Asghar, E. De Cristofaro, D. Kaafar
Private Processing of Outsourced Network Functions: Feasibility and Constructions
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45. L. Onwuzurike and E. De Cristofaro
Experimental Analysis of Popular Smartphone Apps Offering Anonymity, Ephemerality, and End-to-End Encryption
1st NDSS Workshop on Understanding and Enhancing Online Privacy (UEOP 2016)
46. K. Krol, M.S. Rahman, S. Parkin, E. De Cristofaro, E. Vasserman
An Exploratory Study of User Perceptions of Payment Methods in the UK and the US
10th NDSS Workshop on Usable Security (USEC 2016)
47. J. Freudiger, E. De Cristofaro, A. Brito
Controlled Data Sharing for Collaborative Predictive Blacklisting
12th Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2015)
48. M. Nagy, T. Bui, E. De Cristofaro, N. Asokan, J. Ott, A.R. Sadeghi
How Far Removed Are You? Scalable Privacy-Preserving Estimation of Social Path Length with Social PaL
8th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2015)
49. L. Onwuzurike and E. De Cristofaro
Danger is my middle name: Experimenting with SSL Vulnerabilities in Android Apps
8th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2015)
50. E. Ayday, E. De Cristofaro, J.P. Hubaux, G. Tsudik
Whole Genome Sequencing: Revolutionary Medicine or Privacy Nightmare?
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Presented as a poster at USENIX Security 2013 and available on arXiv under the title "The Chills and Thrills of Whole Genome Sequencing"
51. K. Krol, E. Philippou, E. De Cristofaro, M.A. Sasse
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9th NDSS Workshop on Usable Security (USEC 2015)
52. G. Danezis and E. De Cristofaro
Fast and Private Genomic Testing for Disease Susceptibility
13th ACM CCS Workshop on Privacy in the Electronic Society (WPES 2014)
53. I. Bilogrevic, J. Freudiger, E. De Cristofaro, E. Uzun
What's the Gist? Privacy-Preserving Aggregation of User Profiles
19th European Symposium on Research in Computer Security (ESORICS 2014)
54. F. Beato, E. De Cristofaro, K.B. Rasmussen
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12th IEEE Annual Conference on Privacy, Security and Trust (PST 2014)
55. C. Blundo, E. De Cristofaro, P. Gasti
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56. E. De Cristofaro
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57. E. De Cristofaro
An Exploratory Ethnographic Study of Issues and Concerns with Whole Genome Sequencing
8th NDSS Workshop on Usable Security (USEC 2014)
58. E. De Cristofaro, H. Du, J. Freudiger, G. Norcie
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8th NDSS Workshop on Usable Security (USEC 2014)
59. M. Nagy, E. De Cristofaro, A. Dmitrienko, N. Asokan, A.R. Sadeghi
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60. E. De Cristofaro, C. Soriente
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61. E. De Cristofaro, S. Faber, G. Tsudik
Secure Genomic Testing with Size- and Position-Hiding Private Substring Matching
12th ACM CCS Workshop on Privacy in the Electronic Society (WPES 2013)
62. A. Chaabane, E. De Cristofaro, M.A. Kaafar, E. Uzun
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ACM SIGCOMM Computer Communication Review (CCR), 2013
Invited at ACM SIGCOMM 2013 as one of the two "Best CCR Papers."
63. G. Norcie, E. De Cristofaro, V. Bellotti
Bootstrapping Trust in Online Dating: Social Verification of Online Dating Profiles
Financial Cryptography and Data Security (FC) Workshop on Usable Security (USEC 2013)
64. A. Cardenas, R. Chow, E. De Cristofaro
Data Handling in the Smart Grid: Do We Know Enough?
IFIP International Conference on Critical Infrastructure Protection (CIP 2013)
65. E. De Cristofaro, C. Soriente
Participatory Privacy: Enabling Privacy in Participatory Sensing
IEEE Network. Vol. 27, No. 1. January 2013 (Submitted March 2011, Accepted January 2012.)
66. E. De Cristofaro, M. Manulis, B. Poettering
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International Journal of Information Security (IJIS). Vol. 12, No. 1, 2013
67. E. De Cristofaro, C. Soriente, G. Tsudik, A. Williams
Tweeting with Hummingbird: Privacy in Large-Scale Micro-Blogging OSNs
IEEE Data Engineering Bulletin. Vol. 35, Number 4, December 2012
68. E. De Cristofaro, P. Gasti, G. Tsudik
Fast and Private Computation of Cardinality of Set Intersection and Union
11th International Conference on Cryptology and Network Security (CANS 2012)
69. E. De Cristofaro, S. Faber, P. Gasti, Gene Tsudik
GenoDroid: Are Privacy-Preserving Genomic Tests Ready for Prime Time?
International Workshop on Privacy in Electronic Society (WPES 2012)
70. C. Blundo, E. De Cristofaro, P. Gasti
EsPRESSo: Efficient Privacy-Preserving Evaluation of Sample Set Similarity
International Workshop on Data Privacy Management (DPM 2012)
71. E. De Cristofaro and R. Di Pietro
Adversaries and Countermeasures in Privacy-Enhanced Urban Sensing Systems
IEEE Systems Journal, Special Issue on Security and Privacy of Complex Systems, 2012
72. E. De Cristofaro and G. Tsudik
Experimenting with Fast Private Set Intersection
International Conference on Trust and Trustworthy Computing (TRUST 2012)
73. E. De Cristofaro, R. Di Pietro
Preserving Query Privacy in Urban Sensing Systems
International Conference on Distributed Computing and Networking (ICDCN 2012)
74. M. Almishari, E. De Cristofaro, K. El Defrawy, G. Tsudik
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International Journal of Network Security. Vol. 14, No. 6, 2012
75. E. De Cristofaro
Sharing Sensitive Information with Privacy
University of California, Irvine, PhD Dissertation, 2011
76. C. Castelluccia, E. De Cristofaro, A. Francillon, M.A. Kaafar
EphPub: Toward Robust Ephemeral Publishing
IEEE International Conference on Network Protocols (ICNP 2011)
77. E. De Cristofaro, M. Manulis, B. Poettering
Private Discovery of Common Social Contacts
9th International Conference on Applied Cryptography and Network Security (ACNS 2011)
78. E. De Cristofaro, Y. Lu, G. Tsudik
Efficient Techniques for Privacy-Preserving Sharing of Sensitive Information
International Conference on Trust and Trustworthy Computing (TRUST 2011)

79. E. De Cristofaro, C. Soriente
PEPSI: Privacy-Enhanced Participatory Sensing Infrastructure
ACM Conference on Wireless Network Security (WiSec 2011)
80. E. De Cristofaro, A. Durussel, I. Aad
Reclaiming Privacy for Smartphone Applications
IEEE International Conference on Pervasive Computing and Communications (PerCom 2011)
81. G. Ateniese, E. De Cristofaro, G. Tsudik
(If) Size Matters: Size-Hiding Private Set Intersection
IACR International Conference on Practice and Theory of Public Key Cryptography (PKC 2011)
82. E. De Cristofaro, J. Kim, G. Tsudik
Linear-Complexity Private Set Intersection Protocols Secure in Malicious Model
IACR Conference on the Theory and Application of Cryptology and Information Security (Asiacrypt 2010)
83. E. De Cristofaro and J. Kim
Some like it private: Sharing Confidential Information based on Oblivious Authorization
IEEE Security and Privacy, July-August, 2010
84. E. De Cristofaro and G. Tsudik
Practical Private Set Intersection Protocols with Linear Complexity
Financial Cryptography and Data Security (FC 2010)
85. V. Auletta, C. Blundo, A. De Caro, E. De Cristofaro, G. Persiano, I. Visconti
Increasing Privacy Threats in the Cyberspace: the Case of Italian e-Passports
Workshop on Lightweight Cryptography for Resource-Constrained Devices (WLC 2010)
86. E. De Cristofaro, X. Ding, G. Tsudik
Privacy-preserving Querying in Sensor Networks
IEEE International Conference on Computer Communications and Networks (ICCCN 2009)
87. E. De Cristofaro, J.M. Bohli, D. Westhoff
FAIR: Fuzzy-based Aggregation providing In-network Resilience for real-time Wireless Sensor Networks
ACM Conference on Wireless Network Security (WiSec 2009)
88. C. Blundo, E. De Cristofaro, A. Del Sorbo, C. Galdi, G. Persiano
A Distributed Implementation of the Certified Information Access Service
European Symposium on Research in Computer Security (ESORICS 2008)
89. C. Blundo, E. De Cristofaro, C. Galdi, and G. Persiano
Validating Orchestration of Web Services with BPEL and Aggregate Signatures
IEEE European Conference on Web Services (ECOWS 2008)
90. E. De Cristofaro
A Secure and Privacy-Protecting Aggregation Scheme for Sensor Networks
IEEE International Symposium on a World of Wireless Mobile and Multimedia Networks (WoWMoM 2007)
91. C. Blundo and E. De Cristofaro
A Bluetooth-based JXME infrastructure
International Symposium on Distributed Objects, Middleware, and Applications (DOA 2007)
92. V. Auletta, C. Blundo, E. De Cristofaro
A J2ME transparent middleware to support HTTP connections over Bluetooth
International Conference on Systems and Network Communications (ICSNC 2007)
93. V. Auletta, C. Blundo, E. De Cristofaro, G. Raimato
A lightweight framework for Web Services invocation over Bluetooth
IEEE International Conference on Web Services (ICWS 2006)
94. V. Auletta, C. Blundo, E. De Cristofaro, G. Raimato
Performance Evaluation of Web Services Invocation over Bluetooth
ACM Conference on Modeling and Simulation of Wireless and Mobile Systems (MSWiM 2006)

Patents

- Method and Apparatus for Preserving Privacy for Appointment Scheduling.
US Patent US8667062B2. Co-inventors: Imad Aad, Valtteri Niemi, Anthony Durussel, Igor Bilogrevic, Jean-Pierre Hubaux
- Method and apparatus for performing distributed privacy-preserving computations on user locations.
US Patent US8954737B2. Co-inventors: Joan Melia Segui, Rui Zhang, Oliver Brdiczka, Ersin Uzun
- Methods for Selection of Collaborators for Online Threat Mitigation.
US Patent US9817977B2. Co-inventors: Julien Freudiger, Ersin Uzun, Alejandro Brito, Marshall Bern
- Method and Apparatus for Privacy and Trust Enhancing Sharing of Data for Collaborative Analytics.
US Patent US9275237B2. Co-inventors: Julien Freudiger, Ersin Uzun, Alejandro Brito, Marshall Bern

- Portable Proxy For Security Management And Privacy Protection And Method Of Use.
  US Patent US9578062B2. Co-inventors: Julien Freudiger, Ersin Uzun, Golam Sarwar
- Methods for Centralized Privacy-Preserving Collaborative Threat Mitigation.
  US Patent US9477839B2. Co-inventors: Julien Freudiger, Ersin Uzun, Alejandro Brito, Marshall Bern
- Computer-Implemented System And Method For Verifying Online Dating Profiles.
  US Patent Application 20140156750. Co-inventors: Victoria Bellotti and Gregory Norcie
- Privacy-Sensitive Ranking of User Data.
  US Patent Application 20150371059. Co-inventors: Igor Bilogrevic, Julien Freudiger, Ersin Uzun
- Private and Distributed Computation of Probability Density Functions.
  US Patent Application 20150372808. Co-inventors: Igor Bilogrevic, Julien Freudiger, and Ersin Uzun

Professional Service

- **PC/Track/Associate Chair:** NDSS 2027 (Associate Chair), ACM AsiaCCS 2026 (Privacy Track), ACM CCS 2025 (Usability & Measurement Track), TTO 2020 (PC Co-Chair), ACM CCS 2018 (Privacy Track), WWW 2018 (Security & Privacy Track), GenoPri 2018 (PC Co-Chair), GenoPri 2015 (PC Co-Chair), PETS 2014 (PC Co-Chair), PETS 2013 (PC Co-Chair), HotPETs 2012 (PC Co-Chair)
- **PC Member (Top-Tier):** IEEE S&P 2027, SIGMETRICS 2027, ICWSM 2027 (Senior PC), ACM CCS 2026, IEEE S&P 2026, WWW 2026 (Senior PC), ICWSM 2026 (Senior PC), IEEE S&P 2025, USENIX Security 2025, ICWSM 2025 (Senior PC), ACM CCS 2024, IEEE S&P 2023, NDSS 2023, WWW 2023, SIGMETRICS 2023, SIGMETRICS 2022, NDSS 2022, WWW 2022, IEEE S&P 2022, ACM IMC 2022, IEEE S&P 2021, SIGMETRICS 2021, SIGMETRICS 2020, IEEE S&P 2020, WWW 2020, USENIX Security 2020, ACM CCS 2019, NDSS 2020, NDSS 2019, ICWSM 2018, ACM CCS 2017, WWW 2017, NDSS 2017, IEEE S&P 2016, NDSS 2016, IEEE S&P 2015, WWW 2014
- **PC Member (Others):** KDD 2024, PETS 2022 (Senior PC), PETS 2018, ACM HyperText 2018, ICDCS 2018, IEEE EuroS&P 2018, PETS 2017, AsiaCCS 2017, WiSec 2016, PKC 2016, PETS 2016, EuroUSEC 2016, GenoPri 2016, PPML 2016, USEC 2015, WPES 2015, Financial Crypto 2015, WiSec 2014, ICCCN 2014, GenoPri 2014, SESOC 2014, WPES 2014, SESOC 2013, CANS 2013, WiSec 2013, Financial Crypto 2013, CANS 2012, PETS 2012, WiSec 2012, DPM 2012, ICCCN 2012, WPES 2012, CCSW 2012, CANS 2011, ISC 2010
- **Journals:** Associate Editor for the Journal of Cybersecurity (Cryptography and Associated Topics, 2015–2018) and ACM Transactions on Privacy and Security (ACM TOPS, 2021–2026).
- **Steering Committees:** ACM CCSW (2018 – ongoing), PETS (2013–2025)
- **Funding:** NSERC (Canada), NSF SaTC (USA), PRIN (Italy), FWO (Belgium), EPSRC (UK), FWF (Austria)
- **Organization:** Dagstuhl Seminar on "PETs and AI: Privacy Washing and the Need for a PETs Evaluation Framework" (March 9–14, 2025); IEEE EuroS&P 2018, General Co-Chair (April 24–26, 2018); Dagstuhl Seminar on "Cybersafety in Modern Online Social Networks" (September 10–13, 2017)

UC Riverside Service

- **Department (UCR CSE):** Faculty Search Committee, Machine Learning (2023/24 and 2024/25); PI and faculty liaison for the Department of Education's GAANN training grant (student selection, mentoring, and reporting)
- **Academic Senate:** Coordinating Committee for General Education (CCGE), June 2024 – April 2025; Committee on Information Technology (CIT), September 2025 – ongoing; AI Council (AIC), January 2026 – ongoing; Undergraduate Academic Integrity Committee, September 2026 – ongoing
- **Cross-Campus:** Founder and Co-Lead of "LOGOS," a UCR multidisciplinary initiative on modeling and improving online dialogue (involving 20+ members from Computer Science, Psychology, Political Science, etc.)

Teaching

- **Teaching at UCR:** CS 165 Computer Security (Fall 2026), CS 260 Computational Cybersafety (Spring 2026), CS 254 Network Security (Spring 2026), CS 163 Privacy (Winter 2026), CS 163 Privacy (Spring 2025), CS 179F Project in Computer Science–Operating Systems (Winter 2025), CS 260 Privacy Seminar (Spring 2024), CS 254 Network Security (Winter 2024), CS 179F Project in Computer Science–Operating Systems (Fall 2023)
- **Teaching at UCL:** Security (BSc/MEng Computer Science, Winter 2023, Winter 2022), Introduction to Cryptography (MSc InfoSec, Fall 2014, Fall 2015, Fall 2016, Fall 2017), Computer Security II (MSc InfoSec, Winter 2014, Winter 2015, Winter 2016, Winter 2017, Winter 2018, Winter 2019, Winter 2020), Privacy Enhancing Technologies (MSc InfoSec, Winter 2019, Winter 2020)
- **Teaching Assistant:** Network Security (UC Irvine, 2011), Crypto Protocols (University of Salerno, Italy, 2006)
- **Coordinator:** Academic Centre of Excellence in Cyber Security Research (ACE-CSR) Distinguished Seminars Series (UCL, 2014–2020), PARC Privacy Lunch (2012–2013), PARC Security Reading Group (2011–2012)

Mentoring

- **Current Supervision at UCR**: Md. Olid Hasan Bhuiyan, Hyosang Ahn (PhD Advisor); Maya Treves (PhD Co-Advisor, with Michalis Faloutsos)
- **Current Supervision at UCL**: Georgi Ganev (PhD Advisor)
- **Graduated PhD Students** (Main PhD Advisor): Luca Melis (2014–2018, now Research Scientist at Meta), Lucky Onwuzurike (2014–2018, now Cybersecurity Manager at World Association of Nuclear Operators), Apostolos Pyrgelis (2015–2018, now Senior Researcher at RISE Research Institutes of Sweden), Alexandros Mittos (2016–2020, now Head of AI Innovation at Voly Group), Bristena Oprisanu (2017–2021, now Forward Deployed Engineer at Google), Antonis Papasavva (2019–2024, now Cyber Threat Intelligence Analyst at Marsh Re), Mohammad Naseri (2019–2024, now Research Scientist at Flower Labs), Alexandros Efstratiou (2020–2024, now Postdoctoral Researcher at University of Washington), Meenatchi Sundaram Muthu Selva Annamalai (2022–2026, now Postdoctoral Researcher at University of Toronto)
- **Post-doc Alumni**: Vincent Primault (2017–2018), Lukasz Olejnik (2016–2017), Kostantinos Papadamou (2022–2023)
- **Other Mentoring**: Savvas Zannettou (Cyprus University of Technology, de-facto PhD co-advisor, 2016–2019); Angelo Massimo Perillo (University of Salerno, PhD student visitor at UCL, 2017); Kaitai Liang (Aalto University, research visitor at UCL, 2015); PARC summer interns Greg Norcie (Indiana University, 2012 and 2013), Igor Bilogrevic (EPFL, 2013), and Abdelberi Chaabane (INRIA, 2012)
- **PhD Committee**: Hélène Orsini (INRIA Rennes, 2025), Joonas Jälkö (Aalto University, 2023), Sina Sajadmanesh (EPFL, 2023), Federica Granese (Institut Polytechnique de Paris, 2023), César Sabater (INRIA, 2022), Nikolaos Lykousas (University of Piraeus, 2022), Raouf Kerkouche (INRIA, 2021), Ania Piotrowska (UCL, 2020), Tejas Kulkarni (University of Warwick, 2019), Jaroslav Sedenka (Masaryk University, 2019), Marc Roeschlin (University of Oxford, 2018), Ingolf Becker (UCL, 2018), Manu Drijvers (IBM/ETH, 2018), Jean-Louis Raisaro (EPFL, 2018), Saman Feghhi (Trinity College Dublin, 2017), Sharad Kumar (University of Cambridge, 2016), Iacovos Kirlappos (UCL, 2016), Iraklis Leontiadis (EURECOM, 2015), Filipe Beato (KU Leuven, 2015), Lukasz Olejnik (INRIA, 2015), Ginés Dólera Tormo (University of Murcia, 2014), Abdelberi Chaabane (INRIA, 2014), Miguel Malheiros (UCL, 2013)